

Assignment 4: 2.9.5

EE25BTECH11054 - S.Harsha Vardhan Reddy

Question:

$|\mathbf{a}| = 3$, $|\mathbf{b}| = 2\sqrt{3}$, and $\mathbf{a} \cdot \mathbf{b} = 6$, then find the value of $|\mathbf{a} \times \mathbf{b}|$.

Solution:

Given:

$$\|\mathbf{a}\| = 3 \quad (1)$$

$$\|\mathbf{b}\| = 2\sqrt{3} \quad (2)$$

$$\mathbf{a}^\top \mathbf{b} = 6 \quad (3)$$

We know:

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta \quad (4)$$

$$\cos \theta = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \quad (5)$$

Thus

$$(\mathbf{a}^\top \mathbf{b})^2 + (\|\mathbf{a} \times \mathbf{b}\|)^2 = \|\mathbf{a}\|^2 \|\mathbf{b}\|^2 \quad (6)$$

Substituting values

$$\|\mathbf{a} \times \mathbf{b}\| = \sqrt{(3^2) \times (2\sqrt{3})^2 - 6^2} \quad (7)$$

$$\|\mathbf{a} \times \mathbf{b}\| = \sqrt{108 - 36} \quad (8)$$

$$\|\mathbf{a} \times \mathbf{b}\| = \sqrt{72} \quad (9)$$

$$\|\mathbf{a} \times \mathbf{b}\| = 6\sqrt{2} \quad (10)$$

$$(11)$$

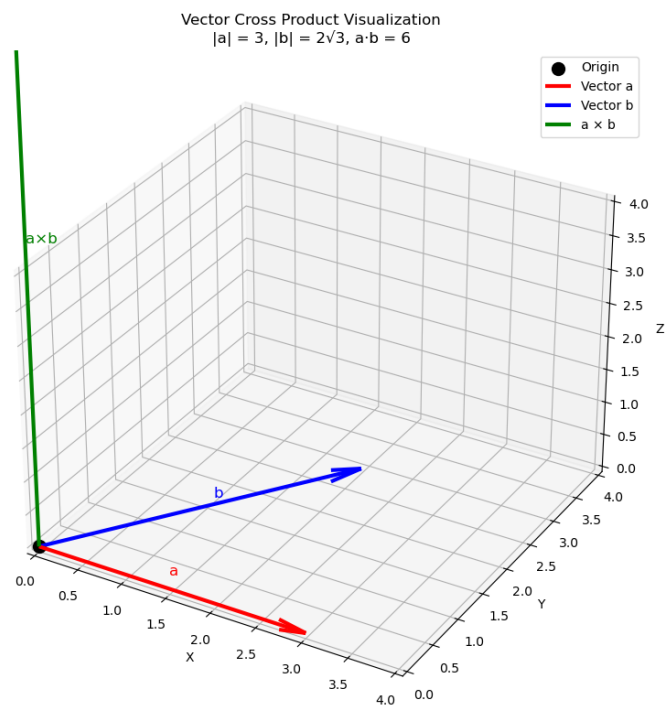


Fig. 0